

WASP-10b

R-2391

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_100916 · created 2026-08-20T23:40:09+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455455.77799 +/- 0.00071 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.179 +/- 0.011
Transit depth (Rp/Rs)^2	3.21 +/- 0.41 [%]
Orbital Inclination (inc)	89.87 +/- 0.48
Airmass coefficient 1 (a1)	1.46 +/- 0.023
Airmass coefficient 2 (a2)	-0.364 +/- 0.073
Scatter in the residuals of the lightcurve fit is	1.59 %
Best Comparison Star	None
Optimal Aperture	3.6
Optimal Annulus	7.46
Transit Duration (day)	0.0966 +/- 0.0011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.179 ± 0.011 vs 0.1592 (deviation 1.58σ)
Duration fit vs published	0.0966 d vs 0.0946 d (deviation 1.61σ)
Mid-time O–C	1.9 min
Depth SNR	14.3
Residual scatter	1.59 %

Light curve

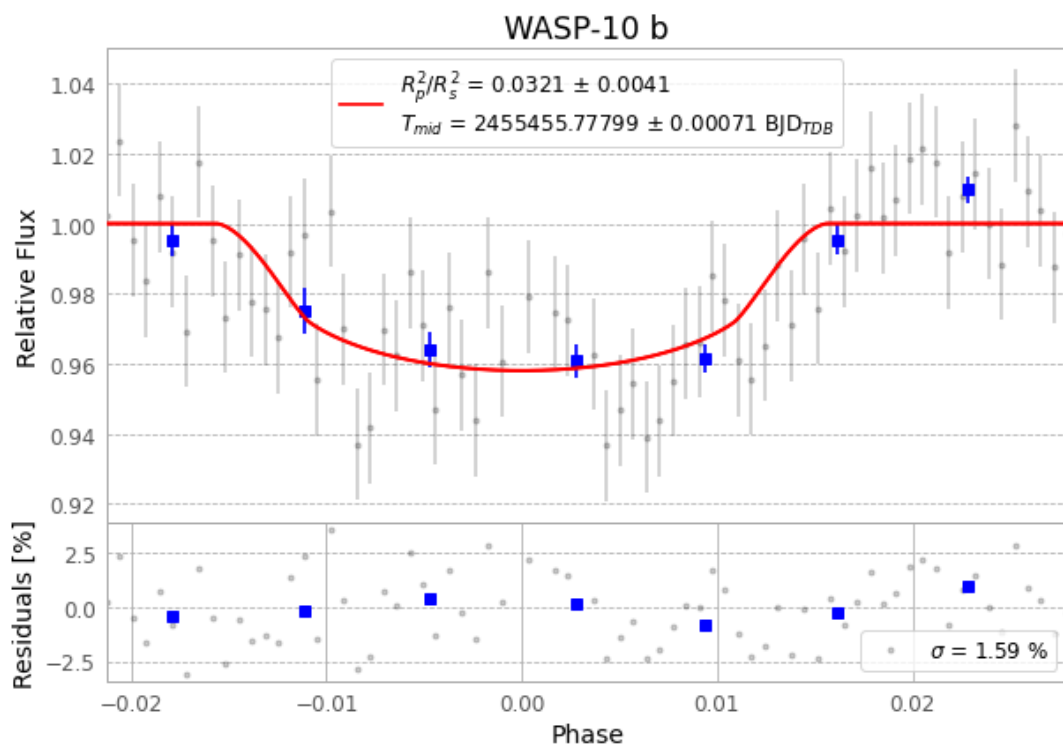


Figure 1 — FinalLightCurve_WASP-10 b_2010-09-15.png (SHA-256 09eb0ce8ee1ab989...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [209, 209], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f6d2a9e79101e534...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

74 FITS frames (49 MB), WASP-10100916045714.FITS → WASP-10100916083614.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-100916.log (6f26ea0733ff...), FinalLightCurve_WASP-10 b_2010-09-15.png (09eb0ce8ee1a...), FinalLightCurve_WASP-10 b_2010-09-15.csv (86e7ef6f64f8...)

Compute resources

Wall time 138.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 23:40Z.